AI-Assisted Health Risk Prediction for Life Insurance Underwriting

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Abstract

Life insurance underwriting relies on estimating an applicant's future mortality risk from health, demographic, and lifestyle information. Traditional actuarial practice favors Generalized Linear Models (GLMs) for their interpretability, yet these may miss complex, non-linear relationships among health factors. I investigate whether modern machine-learning models can improve mortality-risk prediction while preserving human-understandable explanations. Using the National Health and Nutrition Examination Survey (NHANES, 2007–2012 cycles) linked to the CDC public-use mortality file (follow-up through 2019), I build and compare a logistic-regression GLM baseline, an XGBoost model, and a feedforward neural network. I apply SHAP (SHapley Additive exPlanations) to interpret predictions and introduce a lightweight ***agentic* pipeline** in which a locally hosted large language model (Qwen 2.5, via Ollama and LangGraph) transforms model predictions and SHAP explanations into human-readable underwriting summaries. I further extend this layer with **retrieval-augmented generation (RAG)** over a corpus of clinical practice guidelines and open-access literature, so that each narrative is written against retrieved source evidence and cites it. On a held-out test set of 3,716 adults, XGBoost achieved the best performance (AUROC 0.907, PR-AUC 0.663, F1 0.613), narrowly exceeding the interpretable GLM baseline (AUROC 0.904). The small margin suggests transparent models remain competitive for this task, an important finding for a regulated domain that demands explainability. The agentic reporting layer produced faithful summaries that referenced the model's top risk driver in every sampled case, and adding retrieval increased evidential grounding (from zero to 3.0 distinct citations per report on average) without degrading that faithfulness.

Keywords

mortality prediction, life insurance underwriting, explainable AI, SHAP, XGBoost, large language models, agentic AI, retrieval-augmented generation, NHANES

1 Introduction

Life insurance underwriting is fundamentally a healthcare risk-assessment problem: insurers estimate an applicant's probability of death within a policy horizon using health measurements, demographics, and lifestyle indicators. Accurate risk stratification is essential both to price policies fairly and to keep coverage accessible. Errors are costly in two directions: underestimating risk threatens solvency, while overestimating risk unfairly denies or overprices coverage.

The insurance industry has traditionally relied on Generalized Linear Models (GLMs), especially logistic regression, because their coefficients are directly interpretable and defensible to regulators. However, GLMs assume largely additive, linear relationships and may fail to capture interactions among health factors, such as the compounding effect of age combined with elevated blood pressure and impaired kidney function. Modern machine-learning methods such as gradient-boosted trees and neural networks can model these interactions, but they are often treated as "black boxes," which is problematic in a domain where individuals are entitled to an explanation for an adverse decision [6].

This tension between predictive power and interpretability motivates my work. I ask four questions:

(1) Can machine-learning models meaningfully outperform a traditional GLM baseline at mortality-risk prediction on publicly available health data?
(2) Can post-hoc explainability (SHAP) reconcile black-box predictions with the interpretability underwriting requires?
(3) Can a lightweight agentic LLM workflow translate model outputs and SHAP explanations into faithful, human-readable underwriting narratives, without the LLM itself making the mortality prediction?
(4) Does adding retrieved clinical literature to the prompt improve the report's evidential quality without degrading its faithfulness to the model's predictions?

Because proprietary underwriting data (medical records, prescription histories, internal mortality experience) is not publicly available, I use NHANES linked to the CDC mortality file. NHANES is a large, publicly documented, nationally representative health survey with linked mortality outcomes, making it well suited to a *transparent and reproducible* study of AI-assisted risk assessment. My contribution is not a production underwriting model but an end-to-end, reproducible pipeline that demonstrates how prediction, explainability, and natural-language reporting can be combined responsibly.

Summary of findings. Four results follow, and I state them here so that the argument is visible from the outset. First, the accuracy advantage of machine learning over the interpretable baseline is very small: XGBoost reaches 0.907 AUROC against the GLM's 0.904 (Section 4.1), so on this task transparency is nearly free. Second, SHAP and the GLM coefficients agree on the leading risk drivers, which suggests the black-box model learned clinically sensible structure rather than artifacts (Section 4.2). Third, an agent that is permitted to explain but never to predict produces narratives that consistently reference the model's own dominant driver (Section 4.3). Fourth, grounding those narratives in retrieved clinical literature adds citable evidence without degrading their fidelity to the model (Section 4.4), and the way retrieval is wired into the

system, rather than retrieval itself, is what determines whether that grounding can be guaranteed.

2 Related Work

I situate my project against four strands of prior work.

Machine learning for mortality prediction on NHANES. Several studies have used NHANES linked mortality files to predict all-cause mortality, applying gradient-boosted trees and deep networks and reporting strong discrimination (AUROC in the high 0.80s to low 0.90s). These works consistently find age, kidney-function markers, and cardiovascular indicators to be dominant predictors. Related work has shown that machine-learning models can improve cardiovascular and mortality risk prediction over traditional actuarial approaches using routine clinical data [8]. My results are consistent with this literature and extend it by directly contrasting an interpretable GLM against XGBoost [2] and neural-network models under an identical preprocessing and evaluation pipeline on NHANES data [1].

Explainable AI in clinical and actuarial risk models. SHAP [4] has become a standard method for attributing model predictions to input features, offering both global importance and per-instance local explanations. Prior applications in clinical risk modeling use SHAP to audit black-box models and surface clinically plausible drivers, improving expert trust. Tree-specific SHAP methods further enable fast, exact attributions and global insight for gradient-boosted models [5]. A stronger position holds that post-hoc explanation is the wrong remedy altogether: for high-stakes decisions one should use inherently interpretable models rather than explain a black box after the fact, because an explanation is not guaranteed to be faithful to what the model actually computes [6]. My results speak to that debate directly, since the interpretable baseline turns out to be nearly as accurate as the black-box alternatives. I adopt the SHAP approach but take an additional step: rather than presenting SHAP plots to an expert, I feed structured SHAP output to a language model that composes a narrative explanation.

LLM agents for report generation. Large language models have been shown to encode substantial clinical knowledge [7], and the ReAct paradigm [10] lets an LLM interleave reasoning with tool calls to ground its outputs. The recurring risk is *faithfulness*: ensuring generated text reflects the underlying model rather than hallucinating. My design mitigates this by (a) restricting the LLM to explanation, never prediction, and (b) grounding it with tools that return authoritative clinical reference ranges and a fixed risk-category mapping.

Retrieval-augmented generation. RAG [3] conditions generation on passages fetched from an external corpus, reducing reliance on parametric memory and providing citable provenance. Two variants are relevant here. In *traditional* RAG, retrieval is a fixed pipeline stage: it always executes, and the passages are placed in the prompt before generation. In *agentic* RAG, retrieval is exposed as one tool among many and the model decides whether to call it. The distinction is usually framed as an efficiency trade-off, but in a regulated setting it is a governance question: an agent that may silently skip retrieval cannot guarantee that any given report is evidence-backed. I therefore adopt traditional RAG and treat the always-on guarantee as a design requirement rather than an implementation detail.

How my work differs. To my knowledge, prior NHANES mortality studies stop at prediction and SHAP interpretation. I add a reproducible, locally hosted *agentic* reporting layer that converts predictions plus explanations into underwriting narratives; I ground that layer in retrieved clinical guidelines using the model's own SHAP attributions to formulate the search queries; and I explicitly evaluate the trade-off between an interpretable baseline and black-box models on the same footing.

3 Methodology

Figure 1 (page 3) illustrates the end-to-end workflow, from data acquisition through the agentic reporting layer.

3.1 Data

I use NHANES cycles **2007–2008, 2009–2010, and 2011–2012**, linked to the CDC **public-use Linked Mortality File** with follow-up through **December 31, 2019**. Older cycles were chosen deliberately: linking earlier survey participants to the fixed 2019 mortality cutoff yields **7–12 years of follow-up**, producing enough observed deaths for a learnable signal while keeping measurements reasonably modern and comparable. Records were joined on the respondent identifier (SEQN), and the analysis was restricted to eligible adults (age ≥ 18, mortality-eligible), the population relevant to life-insurance underwriting. The prediction target is `MORTSTAT` (1 = deceased by end of follow-up, 0 = alive).

3.2 Features and Preprocessing

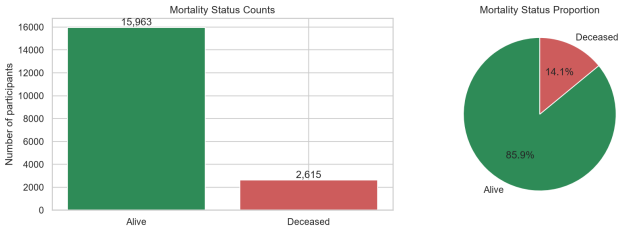


Figure 2: Mortality-outcome class distribution. The deceased class is the minority (≈14%), motivating class-weighted training.

I selected clinically meaningful variables spanning demographics (age, gender, race/ethnicity, education, marital status, poverty-income ratio), body measures (BMI, waist, height, weight), blood pressure, lipids (total and HDL cholesterol), glycemic markers (HbA1c), complete blood count, a biochemistry profile (creatinine, uric acid, albumin, liver enzymes), kidney markers (urine albumin/creatinine), and questionnaire items (smoking, diabetes history, prior cardiovascular events, physical activity). All preprocessing was fit on the **training set only** to prevent leakage:

- **Leakage removal.** Variables known only *after* death (underlying cause of death, death-certificate flags, follow-up time) were excluded.

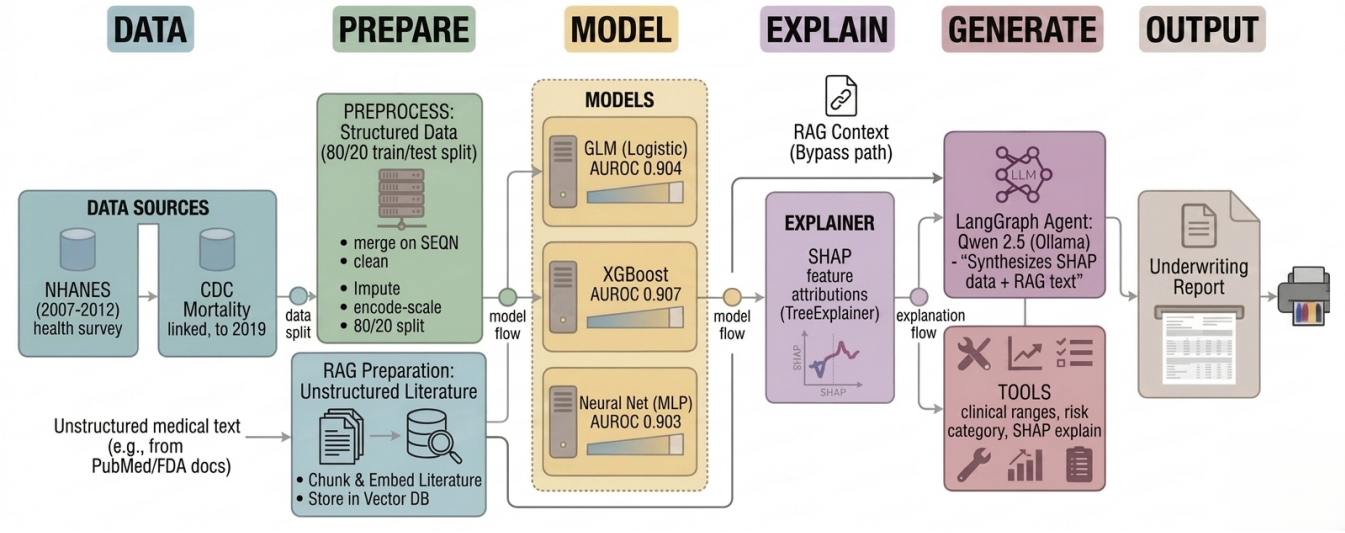


Figure 1: End-to-end architecture of the AI-assisted underwriting pipeline. NHANES survey data and the CDC Linked Mortality File are merged on SEQN, cleaned and feature-engineered, then split before fitting three models. SHAP explains the best model, and its per-applicant attributions serve two purposes: they are passed to the LangGraph agent, and they formulate the queries used to retrieve grounding literature from the ChromaDB vector store. The LLM explains; it never predicts.

- **Missing / sentinel handling.** NHANES refusal / "don't know" codes (7/9, 77/99) were set to missing; continuous features were median-imputed, categorical features mode-imputed.
- **Redundancy.** Fasting glucose was dropped in favor of HbA1c (correlation ≈ 0.77, visible in Figure 3); HbA1c is the more stable clinical marker.
- **Derived features.** Mean systolic/diastolic blood pressure, pulse pressure, a hypertension flag (≥130/80), and BMI category.
- **Skew and scaling.** Right-skewed continuous features were log-transformed (log $1p$); continuous features were standardized and categoricals one-hot encoded.

After a stratified 80/20 split, the training set contained 14,862 records and the test set 3,716, preserving the ≈14% mortality rate in both (Figure 2). The positive rate of 0.141 motivated an imbalance-aware training strategy.

3.3 Models

- **GLM baseline: Logistic Regression.** A Binomial GLM with a logit link, the actuarial standard; class imbalance handled with `class_weight=balanced`.
- **XGBoost.** Gradient-boosted trees with early stopping on validation PR-AUC; imbalance handled via `scale_pos_weight` = 6.10 (the negative/positive ratio).
- **Neural Network.** A feedforward MLP (128→64 hidden units, batch normalization, dropout 0.3) in PyTorch, trained with a class-weighted `BCEWithLogitsLoss` (`pos_weight` = 6.10) and early stopping on validation PR-AUC (Figure 4).

All three models consume the identical preprocessed feature matrix, enabling a fair comparison.

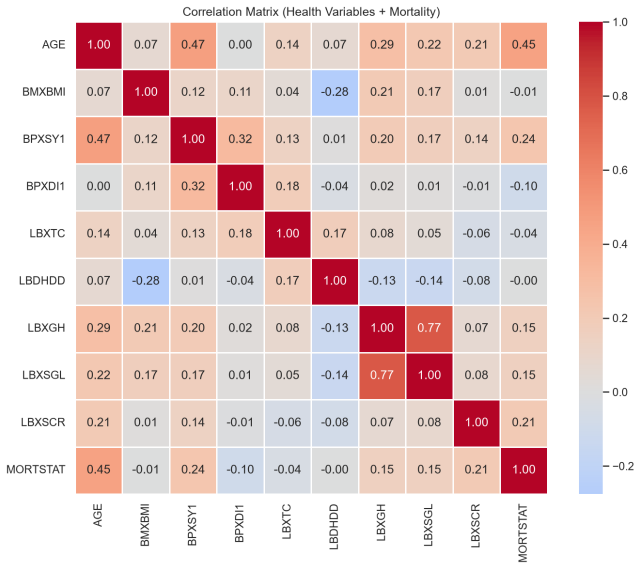


Figure 3: Correlation matrix of continuous features. The high correlation between fasting glucose (LBXSGL) and HbA1c (LBXGH) at ≈0.77 motivated dropping glucose in favor of the more clinically stable HbA1c.

3.4 Explainability

I compute **SHAP** [4] values for the XGBoost model using the exact `TreeExplainer` [5], yielding global importance (mean |SHAP|), dependence relationships, and per-applicant local attributions in

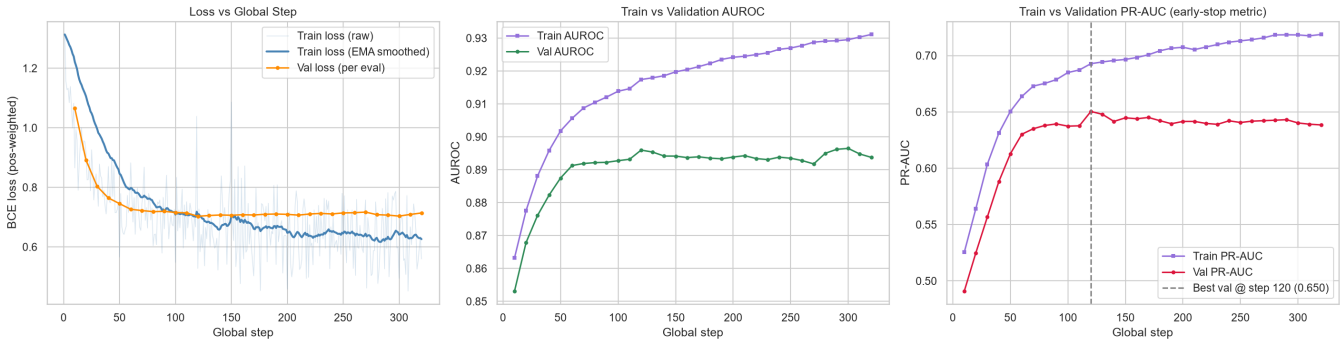


Figure 4: Neural-network training dynamics versus global step: loss (left), train/validation AUROC (center), and train/validation PR-AUC (right, the early-stopping criterion).

log-odds space. SHAP importance is compared against GLM coefficients as a cross-method agreement check.

3.5 Agentic Underwriting Report

The final stage is a **LangGraph ReAct agent** [10] driven by a locally hosted **Qwen 2.5 (7B)** [9] model served through **Ollama**. For each applicant, the agent receives the predicted risk and top SHAP contributions and autonomously decides which of three tools to invoke: (1) `lookup_clinical_ranges`, which interprets a value against published reference ranges (ADA, ACC/AHA, NCEP, WHO) loaded from an auditable JSON configuration; (2) `get_risk_category`, which maps the predicted probability to a simplified underwriting decision (Preferred → Decline); and (3) `explain_shap_contribution`, which renders a SHAP value as a plain-English risk statement. Crucially, **the LLM only explains; it never predicts**. The mortality probability comes solely from XGBoost (best model). Standardized feature values are converted back to original units using the saved scaler statistics before being passed to the agent, so the clinical tool receives real-world values.

3.6 Retrieval-Augmented Grounding

The tool-based agent above grounds *numeric* interpretation (which band a value falls in) but not *clinical reasoning* (what a given combination of findings implies for life expectancy). That knowledge lives in unstructured guidelines and literature, so I add a retrieval stage and evaluate it as a parallel variant of the reporting layer. Both variants share the identical agent and the identical three tools; retrieval is the only difference, which makes the comparison in Section 4.4 a controlled one.

Corpus. I assembled **50 documents**: seven clinical practice guidelines and government statistical reports (ADA *Standards of Care* diagnosis and classification chapter; ACC/AHA hypertension and blood-cholesterol guidelines; KDIGO chronic-kidney-disease guideline, full and summary; NHLBI ATP III lipid reference; CDC/NCHS *United States Life Tables*), the WHO obesity fact sheet, and 42 open-access article full texts retrieved through the Europe PMC REST API across twelve clinical topics chosen to match the model's dominant SHAP drivers (HbA1c, blood pressure, HDL, creatinine and eGFR, BMI, liver enzymes, smoking, heart failure, stroke, cancer survivorship, albuminuria, serum albumin). Acquisition is scripted, so the corpus is reproducible; each retrieved article is stored with its title, journal, authors and DOI for citation.

Indexing. Documents are parsed to text, split into 900-character chunks with 150-character overlap using a recursive splitter that prefers paragraph, then line, then sentence boundaries, and embedded with `nomic-embed-text` (768 dimensions) served locally by Ollama. The resulting **5,593 chunks** are stored in a persistent **ChromaDB** collection under cosine distance. Indexing is a one-off preprocessing cost; inference-time retrieval reuses the persisted index for quick retrieval per applicant across the five queries.

Multi-query retrieval. Rather than embedding the raw applicant record, I derive the queries from the model's own explanation. Each of the applicant's top four SHAP drivers is mapped from its NHANES variable code to a clinical search phrase (for example `MCQ160B` → "congestive heart failure prognosis and mortality"), and one general underwriting query is appended. The five result sets are merged, de-duplicated, ranked by cosine similarity, and the top four passages are injected into the prompt as a numbered block. The system prompt instructs the agent to cite them inline as [1], [2], to cite only numbers present in the block, and to state explicitly when the retrieved literature does not cover a factor.

Traditional, not agentic. Retrieval executes in the orchestration code *before* the language model is invoked; it is not exposed as a tool. This is a deliberate choice. Exposing retrieval as a tool would let a 7B model decide, per applicant, whether to seek evidence, which is acceptable for open-domain question answering but not for a workflow in which every adverse decision must be defensible. The cost is a larger prompt (roughly 450 characters for the tools-only variant versus 3,500 with retrieval) and correspondingly higher latency; the benefit is that grounding is guaranteed by construction rather than left to sampling.

3.7 Evaluation

Models are evaluated on the held-out test set with **AUROC** (discrimination), **PR-AUC** (average precision, suited to imbalance), and **F1** for the positive (deceased) class (Figure 6 visualizes the first two). For the generative layer I apply an automated *faithfulness* check: whether each report references the model's top SHAP driver. Matching is *synonym-expanded*, because a check on the raw variable code alone is too strict: the agent writes in clinical prose,

Table 1: Test-set performance (n = 3,716). Best value per column in bold.

Model	AUROC	PR-AUC	F1
XGBoost	**0.907**	**0.663**	**0.613**
Neural Network	0.903	0.653	0.560
GLM (Logistic Reg.)	0.904	0.649	0.579

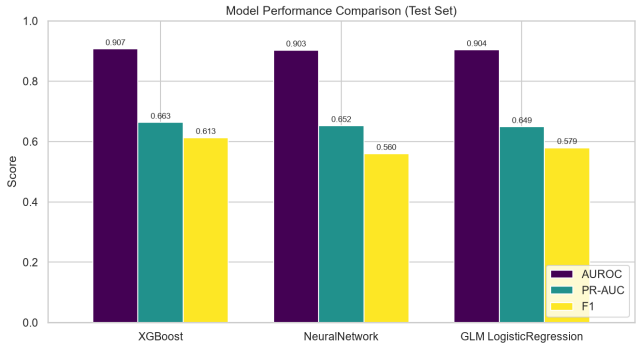


Figure 5: Model performance comparison across AUROC, PR-AUC, and F1.

so `BMXWAIST` legitimately appears as "waist circumference" and `MCQ160B` as "congestive heart failure". A single synonym table is shared by the reporting and consolidation stages, and both arms of the tools-only versus RAG comparison are scored with it, so the figure is consistent throughout.

For the retrieval variant I additionally record *citation density* (distinct bracketed references per report), the number of passages injected, the mean cosine similarity of retrieved passages, report length, and generation latency. I stress that citation density measures *coverage*, not correctness: it confirms the model used the supplied evidence, not that it used it correctly.

4 Results

4.1 Model Comparison

Table 1 reports test-set performance for all three models. XGBoost achieved the best scores on every metric, but its margin over the interpretable GLM baseline is small (AUROC +0.003, PR-AUC +0.014). The neural network performed comparably. Figure 5 visualizes the comparison, and Figure 6 overlays the ROC and precision–recall curves.

The central finding is that a transparent GLM is *highly competitive* with black-box models on this tabular clinical data. For a regulated domain that requires explainable adverse decisions, the marginal predictive gain of XGBoost may not justify the loss of native interpretability, though, as I show next, SHAP largely recovers it.

4.2 Risk Drivers and Explainability

Age was by far the strongest predictor in both the GLM and the SHAP analysis. In the GLM, a one-standard-deviation increase in age multiplied the odds of death by $e^{1.47} \approx 4.3$ (Table 2, row 1). SHAP assigned age a mean |SHAP| of 1.41, several times larger than any other feature (Figure 7). Renal and metabolic markers (urine albumin, creatinine), socioeconomic status (poverty-income ratio), smoking, and prior cardiovascular events followed. Figure 8 resolves the same attributions to the level of individual applicants and shows the *direction* of each effect: high values of age and urine albumin push predicted mortality upward, while a higher poverty-income ratio pushes it down, which is the expected socioeconomic gradient. Table 2 shows strong agreement between the two methods on the leading drivers, which increases confidence that the models learned clinically sensible relationships rather than artifacts.

Table 2: Top-ranked mortality risk drivers: SHAP (XGBoost) vs. GLM coefficients. Both methods rank age first. GLM coefficients are standardized (one-SD change); the odds ratio e^{β} gives the multiplicative change in odds per SD.

#	SHAP driver	$\overline{\lvert\text{SHAP}\rvert}$	GLM driver	β	odds Ratio
1	Age	1.41	Age	+1.47	4.33
2	Urine albumin	0.27	Race/ethnicity	+0.53	1.70
3	Poverty ratio	0.19	BMI	−0.45	0.64
4	Race/ethnicity	0.16	CHF (yes)	+0.38	1.46
5	Smoking (ever)	0.14	Marital status	−0.39	0.68

Notably, BMI carried a *negative* GLM coefficient (higher BMI associated with lower predicted mortality). This counter-intuitive result reflects reverse causality, since serious illness often causes weight loss. It is a well-documented epidemiological artifact rather than evidence that higher BMI is protective. This is exactly the kind of subtlety that an explainability layer surfaces for human review.

4.3 Agentic Underwriting Reports

The LangGraph agent produced concise, structured underwriting summaries for sampled applicants. Figure 9 shows the SHAP local explanation underlying a high-risk case, and the box below shows the corresponding agent-generated report.

Example agent-generated report (Applicant 118)

Predicted risk 99.4% (actual: deceased). *Decision:* Decline; refer for medical review. *Key factors:* advanced age (80 years, +2.23 log-odds); prior cardiovascular conditions; low HDL cholesterol; mildly elevated creatinine. *Summary:* risk is extremely high, driven primarily by advanced age with additional renal and lipid contributions.

To assess whether reports faithfully reflect the model, I checked whether each report referenced the applicant's top SHAP driver, allowing clinical synonyms. **All 8 of 8** sampled reports did, indicating the generated narratives are grounded in the model's actual reasoning rather than hallucinated.

4.4 Tools-Only versus Retrieval-Augmented Reports

Table 3 compares the two reporting variants on the same eight applicants, using the same prediction and the same SHAP values. Because the agent and its three tools are identical across arms, the differences are attributable to retrieval alone.

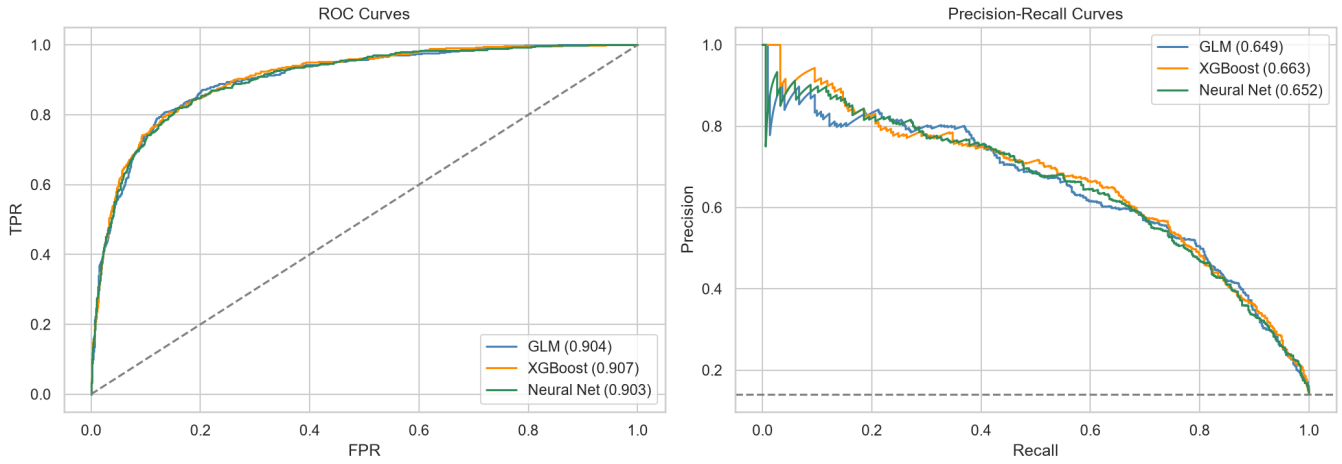


Figure 6: ROC (left) and precision–recall (right) curves for all three models on the test set.

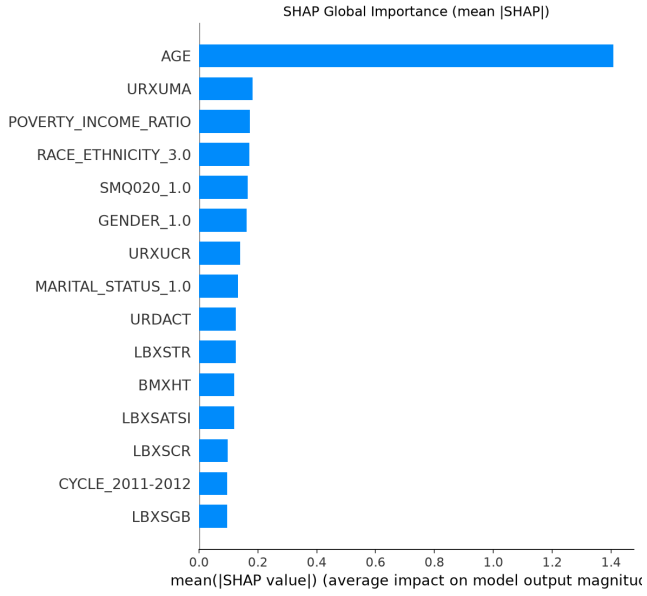


Figure 7: Global SHAP feature importance (mean |SHAP|) for XGBoost. Age dominates the model's predictions.

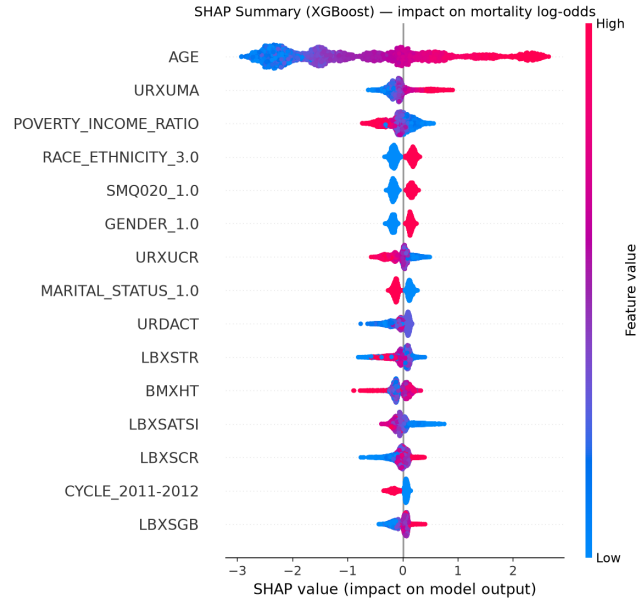


Figure 8: SHAP beeswarm summary. Each point is one applicant; horizontal position is the feature's impact on predicted mortality log-odds, colored by feature value.

Three observations follow. First, retrieval delivered what it was added for: narratives moved from zero external references to 3.0 distinct citations each, and reports grew by roughly a third (179 to 242 words), the additional text being the literature-grounded clinical context. Retrieved passages had a mean cosine similarity of 0.715, topically relevant without being near-duplicates of the query. Second, and importantly for a safety argument, *faithfulness did not deteriorate*: both arms name the top SHAP driver in all eight reports, so the additional context did not displace the SHAP-derived narrative or dilute the model's reasoning. Because the baseline is already at ceiling on this metric, the comparison can only demonstrate non-degradation, not improvement. Third, the dominant cost is prompt size: the retrieval block makes the prompt roughly eight times larger, and generation latency averaged 88.5 s per report (range 47–127 s) on CPU. Latency for the tools-only arm was not instrumented, so I do not report a ratio.

4.5 Demonstrator Application

To show the pipeline in operational use rather than as a batch experiment, I built a local web application (Gradio) that executes the full chain for a single applicant. Figure 10 shows its control flow. The underwriter either selects a held-out test-set patient or enters values manually, and chooses the reporting approach (tools-only,

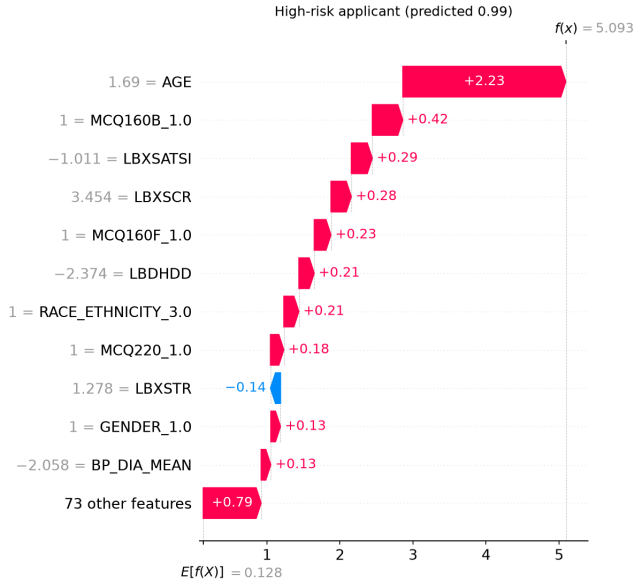


Figure 9: SHAP waterfall for a high-risk applicant (predicted risk 99.4%, actually deceased). Age is the dominant upward contributor.

Table 3: Reporting-layer comparison (n = 8 applicants). Both arms use the same agent, the same three tools, and the same SHAP inputs; only retrieval differs. Faithfulness uses the synonym-expanded matcher for both arms.

Metric	Tools only	Tools + RAG
Passages injected per report	0	4
Distinct citations per report (mean)	0.0	3.0
Mean retrieval similarity (cosine)	n/a	0.715
Report length (mean words)	179	242
Prompt size (characters)	~450	~3,500
Top SHAP driver named	8/8	8/8
Generation latency (mean seconds)	n/a	88.5

tools plus RAG, or both rendered side by side) *before* running the assessment. Feature assembly fills unspecified variables from training medians, the saved preprocessor and XGBoost model produce the risk estimate, and `TreeExplainer` computes local attributions. If the selected approach includes retrieval, the SHAP drivers are converted into search queries at that point and the matched passages are displayed (query, source file, page and similarity score) before the agent is invoked. The activity log streams each stage, tool call and elapsed time as it happens.

Exposing retrieval in the interface is a deliberate transparency measure. An underwriter can see which passages the narrative was written against and judge their relevance, which is a precondition for the human-in-the-loop review that this kind of system requires.

Table 4: Principal findings against the four research questions of Section 1, and their practical implications.

Question	Finding	Implication
RQ1: ML vs. GLM	XGBoost 0.907 AUROC vs. GLM 0.904; PR-AUC +0.014	XGBoost achieves the best accuracy with only a marginal gain over the GLM, so a regulated underwriter can prefer XGBoost paired with SHAP for the slight edge while retaining full explainability
RQ2: SHAP recovers interpretability	SHAP and GLM agree on age as dominant and on several secondary markers	Post-hoc attribution is a viable audit route for a black-box model, and cross-method agreement is a practical sanity check
RQ3: Agentic narrative	8/8 reports named the model's top SHAP driver	Separating prediction from explanation yields narratives that track the model rather than inventing rationales
RQ4: Retrieval grounding	0 → 3.0 citations per report; faithfulness unchanged at 8/8	Evidence can be added without displacing the model's own reasoning

4.6 Summary of Principal Findings and Their Implications

Table 4 maps each research question to its result and to what that result implies for practice. Section 5 develops the two most consequential implications in detail.

5 Discussion

XGBoost wins, and SHAP keeps it explainable. The near-tie the accuracy comparison between the GLM and XGBoost is shown in Table 1 for this tabular clinical task. In life-insurance underwriting, where adverse decisions must be explained and are subject to regulatory scrutiny, this means a practitioner can choose XGBoost for its slight predictive edge and recover explainability through SHAP, rather than sacrificing accuracy by falling back to a GLM. Rudin [6] argues that high-stakes applications should prefer inherently interpretable models, and my results show the cost of that preference is genuinely small. But they also show there is an alternative: pairing a black-box model with faithful post-hoc explanation (SHAP plus the agentic narrative) achieves both the best accuracy and a defensible explanation at the cost of a more complex pipeline.

Explainability plus generation. SHAP recovered interpretability for XGBoost, and the agentic layer translated numeric attributions into readable underwriting language while keeping prediction and explanation strictly separated. This separation of concerns is a practical template for responsible AI assistance: the model decides, the explanation is auditable, and the narrative is grounded.

Grounding is an architectural property, not a prompt. The most transferable lesson from Section 3.6 is where the retrieval decision is placed. Tool-based (agentic) RAG is attractive because the

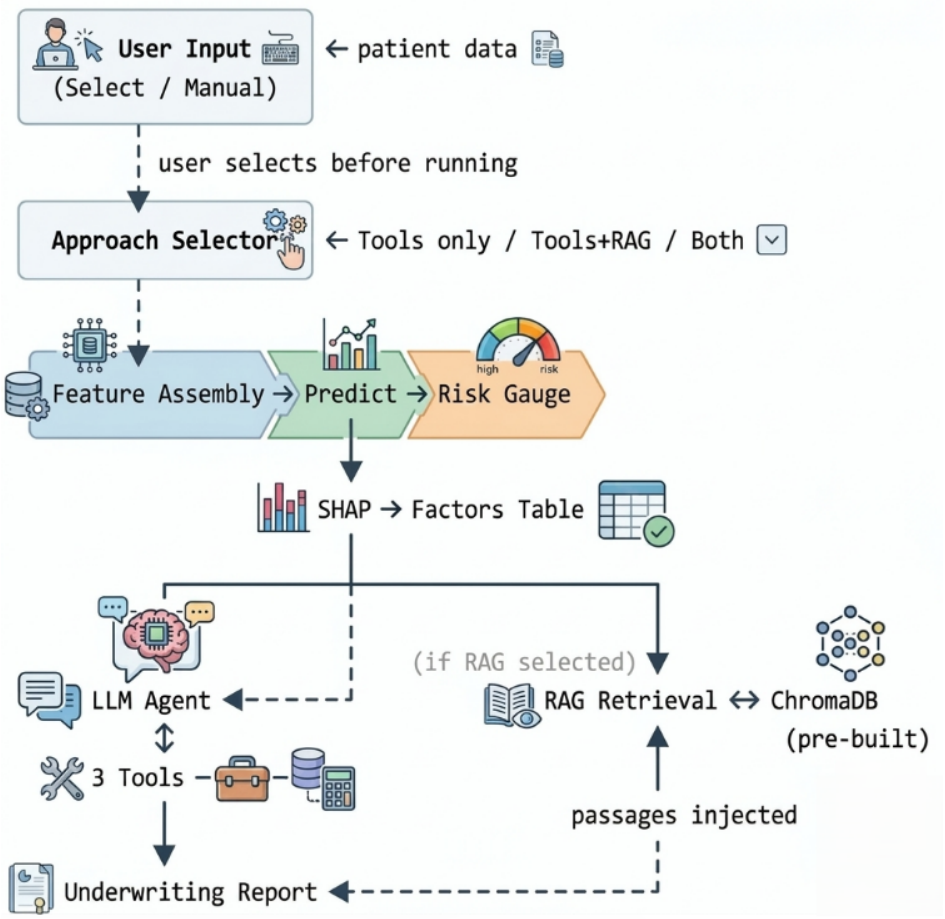


Figure 10: Control flow of the demonstrator application. The reporting approach is an up-front user choice that gates the retrieval branch; retrieval itself runs after SHAP, because the search queries are derived from the applicant's SHAP drivers.

model fetches evidence only when it judges it necessary, but that judgement is exactly what cannot be delegated in a regulated workflow: a report that silently skipped retrieval is indistinguishable, in the output artifact, from one that retrieved and found nothing. Moving retrieval into the orchestration code converts a probabilistic behaviour into an invariant. The same reasoning motivates surfacing the retrieved passages in the interface: provenance that is recorded but not visible cannot support review. I regard the extra prompt tokens and latency in Table 3 as the price of that invariant, and a low one.

A related caution: because the search queries are built from the model's own SHAP drivers, the retrieved literature will always match what the model considered important. That makes the results relevant, but it also means that if the model emphasised the wrong factor, the retrieved evidence would still back that wrong emphasis. In other words, RAG makes the *report* evidence-backed, but it does not check whether the *model itself* is correct.

Limitations. (1) NHANES is not proprietary underwriting data; real insurers use richer medical and prescription histories, so absolute performance may not transfer. (2) Treating mortality as a binary label ignores time-to-event; a survival model (e.g., Cox proportional hazards) would exploit the follow-up months more fully. (3) Despite class weighting, PR-AUC (≈0.66) shows precision/recall trade-offs remain for the minority (deceased) class. (4) Reverse causality (e.g., low BMI) can mislead naive interpretation. (5) The local LLM's output varies and requires human review; the risk-category thresholds are illustrative, not actuarially calibrated. In particular, the 7B model occasionally mis-sorted a risk-increasing factor into the "protective factors" heading, a formatting error that a reviewer would catch but an automated metric does not. (6) Report quality is bounded by the corpus. My 50 documents cover only the model's dominant drivers. Retrieval is also unevenly distributed across the corpus: the life-table and kidney-guideline documents supplied over half of all injected passages, reflecting the dominance of age and renal markers among the model's SHAP drivers rather than balanced coverage of the literature. (7) Citation density measures coverage, not correctness. Verifying that each cited passage actually supports the sentence it is attached to requires expert adjudication that I did not perform. (8) The reporting comparison covers only eight

applicants, which is too few to draw statistically confident conclusions. Additionally, both approaches already scored a perfect 8/8 on faithfulness, so there is no room to detect improvement, only deterioration.

6 Conclusion and Future Directions

I presented a reproducible, end-to-end pipeline for AI-assisted mortality-risk assessment in life-insurance underwriting, combining a GLM baseline, XGBoost, and a neural network with SHAP explainability and an agentic LLM reporting layer. On NHANES data linked to CDC mortality, XGBoost was best (AUROC 0.907) but only narrowly ahead of an interpretable GLM (AUROC 0.904), and SHAP plus a grounded LLM agent produced faithful, human-readable underwriting summaries.

Extending that layer with retrieval-augmented generation over 50 clinical guideline and literature sources (5,593 indexed passages) moved reports from zero external references to 3.0 citations each without degrading their faithfulness to the model's attributions. I argue that the placement of the retrieval decision matters as much as retrieval itself: implementing RAG as a deterministic pipeline stage rather than an optional agent tool makes evidential grounding an invariant of the system, which is the property a regulated workflow actually needs. A demonstrator application exposes the whole chain, including the retrieved passages, for human review.

6.1 What I Would Do Differently

With hindsight, there are four decisions I would make differently if I repeated this project.

Frame the target as time-to-event from the start. Collapsing the CDC follow-up months into a binary `MORTSTAT` label discarded the most informative part of the outcome. Two applicants both coded "deceased" are treated identically whether death occurred at month 6 or month 140, yet those are entirely different underwriting risks, and policy pricing is inherently a function of horizon. A Cox proportional-hazards or discrete-time survival formulation was available from the same data at no extra collection cost. This was the single largest missed opportunity, and it was a framing choice made early that constrained everything downstream.

Evaluate retrieval quality before generating reports. I added RAG after the agent was already working, so my first indication of whether the retrieved passages were useful came from reading the final reports. A better approach would have been to evaluate retrieval in isolation first: check whether the top-ranked passages actually contain clinically relevant information for each SHAP driver, and tune the chunk size, overlap and number of queries before ever passing the results to the LLM.

Budget the context window as a design parameter. The most costly practical failure of the project was not statistical but architectural: the retrieval-augmented prompt overflowed the model server's default context, which manifested as a silent stall rather than an error and cost significant debugging time. Treating tokens-per-turn as an explicit budget, and configuring an explicit request timeout, from the first agentic experiment would have surfaced this immediately. I would also instrument latency for every arm from the outset; the tools-only baseline was never timed, so Table 3 cannot report a cost ratio.

Define report-quality metrics before generating reports. Faithfulness was operationalised after the reports had been produced, and the score proved sensitive to how the matcher treats clinical synonyms: a variable code such as `BMXWAIST` must be recognised in the phrase "waist circumference" for the check to be meaningful. Settling that definition in advance, and validating it against a small hand-labelled set, would have made the claim considerably stronger than choosing the rule after seeing the text. Relatedly, eight applicants is too few; sampling a stratified fifty would have cost only additional compute time.

6.2 Future Directions

Beyond those corrections, I see five promising directions. (i) **Survival modelling** using the follow-up duration, which would let the system express risk as a horizon-specific curve rather than a single probability, much closer to how policies are actually priced. (ii) **Calibration and fairness auditing** across demographic groups. Discrimination (AUROC) is not sufficient for pricing, which needs calibrated probabilities; and because race, sex and socioeconomic status appear among the model's drivers, subgroup calibration and disparate-impact analysis are ethical prerequisites rather than extensions. (iii) **A controlled traditional-versus-agentic RAG comparison** on the same corpus, measuring how often a tool-based agent declines to retrieve and what that omission costs in evidential quality. My design argument for deterministic retrieval is currently reasoned rather than measured. (iv) **Expert adjudication of citation correctness**, replacing citation density with a judgement of whether each cited passage actually supports the sentence it is attached to. That is the difference between grounding that looks rigorous and grounding that actually is. (v) **A human evaluation with practising underwriters**, assessing whether these narratives change decision quality, speed, or confidence relative to reading SHAP plots directly. That is ultimately the only test that matters for a decision-support tool.

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